Honey bees suffer from an infectious disease called foul brood. This attacks the grubs in their cells. Of the domestic breeds used by beekeepers, some are more at risk from foul brood than others, and it turns out that the difference between strains is, at least in some cases, a behavioural one. There are so-called hygienic strains which quickly stamp out epidemics by locating infected grubs, pulling them from their cells and throwing them out of the hive. The susceptible strains are susceptible because they do not practise this hygienic infanticide. The behaviour actually involved in hygiene is quite complicated. The workers have to locate the cell of each diseased grub, remove the wax cap from the cell, pull out the larva, drag it through the door of the hive, and throw it on the rubbish tip.

Doing genetic experiments with bees is quite a complicated business for various reasons. Worker bees themselves do not ordinarily reproduce, and so you have to cross a queen of one strain with a drone (= male) of the other, and then look at the behaviour of the daughter workers. This is what W. C. Rothenbuhler did. He found that all first-generation hybrid daughter hives were non-hygienic: the behaviour of their hygienic parent seemed to have been lost, although as things turned out the hygienic genes were still there but were recessive, like human genes for blue eyes. When Rothenbuhler 'back-crossed' first-generation hybrids with a pure hygienic strain (again of course using queens and drones), he obtained a most beautiful result. The daughter hives fell into three groups. One group showed perfect hygienic behaviour, a second showed no hygienic behaviour at all, and the third went half way. This last group uncapped the wax cells of diseased grubs, but they did not follow through and throw out the larvae. Rothenbuhler surmised that there might be two separate genes, one gene for uncapping, and one gene for throwing-out. Normal hygienic strains possess both genes, susceptible strains possess the alleles—rivals—of both genes instead. The hybrids who only went half way presumably possessed the uncapping gene (in double dose) but not the throwing-out gene. Rothenbuhler guessed that his experimental group of apparently totally non-hygienic bees might conceal a sub-group possessing the throwing-out gene, but unable to show it because they lacked the uncapping gene. He confirmed this most elegantly by removing caps himself. Sure enough, half of the apparently non-hygienic bees thereupon showed perfectly normal throwing-out behaviour.*